

A Multiscale Modeling Framework for Contrail Formation and Persistence Using Plume-Scale Simulations and Reanalysis Meteorology

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1 Introduction

1.1 Contrails and Radiative Forcing

Aircraft condensation trails (contrails) are line-shaped ice clouds that form in the upper troposphere when moist aircraft exhaust mixes with cold ambient air and becomes supersaturated with respect to ice (2; 6). Under favorable atmospheric conditions, contrails can persist for hours and spread into cirrus-like cloud decks (contrail cirrus), exerting a net warming influence on Earth’s climate (3; 4; 5). Unlike low-level clouds, which primarily affect shortwave radiation, contrails typically form at high altitudes where temperatures are substantially lower than at the surface, making their longwave radiative impact dominant (4; 5).

The physical basis for this effect follows directly from the Stefan–Boltzmann law,

$$F = \sigma T^4, \quad (1)$$

which governs thermal emission from a body at temperature T . Because emission scales strongly with temperature, radiation emitted from cold upper-tropospheric cloud layers is substantially weaker than radiation emitted from the warm surface and lower troposphere.

Figure 1 illustrates this mechanism schematically. A contrail forming near the tropopause acts as a radiatively cold layer overlying a much warmer atmosphere below. Longwave radiation emitted to space is therefore characteristic of the contrail temperature rather than the surface temperature, leading to a reduction in outgoing longwave radiation and a positive radiative forcing (3; 4; 5).

1.2 Role of Vertical Temperature Structure

The sign and magnitude of contrail radiative forcing depend critically on the vertical temperature profile of the atmosphere. In an idealized two-layer framework consisting of a surface at temperature T_s and an overlying contrail layer at temperature T_c , the clear-sky outgoing longwave radiation may be approximated as

$$F_{\text{clear}} \approx \sigma T_s^4, \quad (2)$$

whereas in the presence of an optically thick contrail, the effective emission to space is instead governed by

$$F_{\text{contrail}} \approx \sigma T_c^4. \quad (3)$$

Since $T_c \ll T_s$ in the upper troposphere, it follows that

$$F_{\text{contrail}} < F_{\text{clear}}, \quad (4)$$

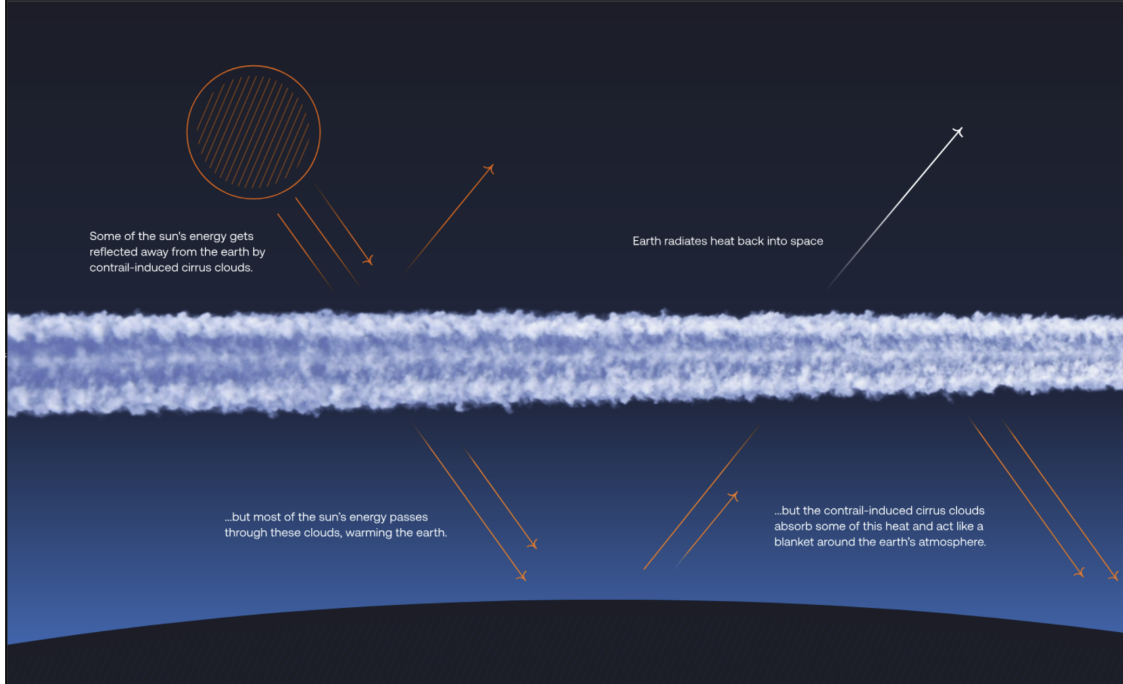


Figure 1: Conceptual illustration of contrail radiative forcing. A contrail forming in the cold upper troposphere overlies a warmer lower atmosphere and surface. Because longwave emission scales as σT^4 , emission to space from the contrail layer is substantially weaker than surface emission, leading to a net longwave warming effect. The schematic emphasizes the role of the vertical temperature profile rather than providing a quantitative radiative transfer calculation.

yielding a positive longwave radiative forcing. This simple derivation highlights that contrail impacts are fundamentally tied to atmospheric thermal structure rather than cloud coverage alone.

Although radiative transfer is not explicitly simulated in this study, the temperature contrast embodied in Figure 1 motivates a focus on upper-tropospheric environments where contrails are both thermodynamically favored and radiatively effective.

1.3 Environmental Controls on Contrail Persistence

Whether a contrail forms and persists depends sensitively on the surrounding atmospheric environment. Persistent contrails are observed primarily in ice-supersaturated regions where relative humidity with respect to ice (RH_i) exceeds 100%, allowing ice crystals to grow rather than sublimate (2; 6; 7). Static stability, commonly quantified by the Brunt–Väisälä frequency N , regulates vertical mixing and plume dilution, while vertical wind shear controls lateral stretching and deformation of the exhaust plume (6; 15).

These controls operate at spatial scales that are often unresolved in global atmospheric models. As a result, plume-scale mixing and microphysical decay must be parameterized, introducing uncertainty into estimates of contrail persistence and radiative forcing.

1.4 Atmospheric Data and Contrail Modeling Framework

To characterize the atmospheric environments associated with contrail formation, this study uses ERA5 reanalysis data (9) together with the `pycontrails` modeling framework (10). ERA5 pressure-

level temperature, humidity, and wind fields are used to compute relative humidity over ice, Brunt–Väisälä frequency, and vertical wind shear in the upper troposphere (9; 16). These fields define the thermodynamic and dynamic background in which contrail formation is expected to occur.

The `pycontrails` framework is used to generate gridded contrail diagnostics and to collocate atmospheric conditions with contrail-relevant regions and times. Rather than relying on satellite-based detection, this approach enables systematic and physically consistent exploration of contrail environments across space and time using reanalysis-driven inputs.

1.5 Hierarchical Modeling Approach

To bridge the gap between large-scale atmospheric conditions and unresolved plume-scale processes, this work adopts a hierarchical computational approach. First, a reduced-order one-dimensional plume model implemented in MATLAB represents the cross-plume evolution of ice mass using diffusion and bulk microphysical tendencies. This model enables systematic parameter sweeps in RH_i , N , and shear S , the construction of phase diagrams, and the derivation of empirical scaling laws for contrail lifetime.

Second, high-fidelity two-dimensional simulations performed in COMSOL provide spatially resolved reference solutions. These simulations explicitly resolve plume advection, shear-driven deformation, and diffusion, allowing direct comparison with reduced-order diagnostics such as centerline decay and effective plume width.

ERA5-driven atmospheric conditions and `pycontrails`-based diagnostics anchor both modeling components in realistic upper-tropospheric environments, linking plume-scale behavior to meteorologically consistent contrail formation regimes.

1.6 Objectives and Scope

The objective of this study is to quantify how relative humidity over ice, static stability, and vertical wind shear jointly control contrail plume persistence. By combining ERA5 reanalysis, `pycontrails`-based atmospheric diagnostics, reduced-order MATLAB modeling, and high-fidelity COMSOL simulations, the study seeks to identify controlling regimes, derive physically interpretable scaling relationships, and clarify the plume-scale processes most relevant for contrail persistence in the atmosphere.

2 Reduced-Order Contrail Plume Model (MATLAB)

This section presents a reduced-order contrail plume model implemented in MATLAB to isolate how three environmental controls, relative humidity with respect to ice (RH_i), static stability (Brunt–Väisälä frequency N), and vertical wind shear, S govern contrail persistence. The code is given at the end of the document.

2.1 Modeling choices and scope

Contrail evolution couples turbulence-driven mixing and dilution, thermodynamic control through supersaturation, and ice microphysics. Fully resolving these processes in three dimensions is expensive and typically requires advanced microphysics and turbulence closures. For this simulation, a reduced model is appropriate because the primary objective is to identify *regimes* and *sensitivity* to environmental parameters that can later be compared against reanalysis-derived distributions and refined using higher-fidelity simulation.

We therefore model a contrail cross-section through a one-dimensional scalar field $q_i(y, t)$, interpreted as a bulk *ice mixing ratio proxy* (kg/kg) as a function of cross-plume coordinate y and time t . Streamwise structure is not explicitly represented at this level. Instead, the model focuses on the competition between:

- **Cross-plume dilution/spreading** represented by an effective diffusion term.
- **Bulk microphysical growth/decay** represented by an RHi-dependent source term.

These two processes are sufficient to generate an interpretable persistence metric τ_{persist} and to construct phase diagrams in (RHi, N, S) .

2.2 Governing equation

The reduced model solves an advection-free diffusion–reaction equation for $q_i(y, t)$:

$$\frac{\partial q_i}{\partial t} = K(N, S) \frac{\partial^2 q_i}{\partial y^2} + \mathcal{M}(q_i; \text{RHi}), \quad y \in \left[-\frac{L_y}{2}, \frac{L_y}{2}\right], \quad t \geq 0. \quad (5)$$

The diffusion term represents the net effect of turbulent mixing and shear-enhanced dispersion as an *effective* cross-plume diffusivity K . The microphysical term $\mathcal{M}$ represents net sublimation when air is subsaturated with respect to ice and net growth (weakly) when supersaturated. This structure is intentionally minimal: it preserves the correct qualitative threshold behavior at $\text{RHi} \approx 100\%$ while remaining simple enough for regime mapping and parametric fitting. The diffusion form and Gaussian spreading behavior follow classical results for scalar diffusion and plume width growth (12; 14), while shear-enhanced dispersion motivates treating shear as an effective mixing amplifier in reduced closures (13).

2.3 Effective diffusivity closure $K(N, S)$

The model parameterizes mixing strength via an effective diffusivity that depends on stability N and shear S :

$$K(N, S) = K_0 \left(1 + k_S \frac{|S|}{S_{\text{ref}}}\right) \left(1 + k_N \frac{N_{\text{ref}}}{\max(N, N_{\text{min}})}\right). \quad (6)$$

This choice is based on two modeling decisions:

- **Shear sensitivity:** A larger velocity gradient increases deformation and interfacial area, enhancing entrainment and effective lateral spreading. The factor $(1 + k_S |S|/S_{\text{ref}})$ introduces a monotone increase of K with $|S|$.
- **Stability sensitivity:** Static stability modulates vertical motions and mixing pathways. Rather than resolving stratified turbulence explicitly, N is treated as an environmental control knob that shifts effective mixing rates over realistic ranges (15; 16). The inverse dependence N_{ref}/N is deliberately simple, algebraically convenient for sweeps, and subject to later validation/calibration against COMSOL (where shear-driven deformation and transport are resolved spatially).

The parameters used in the model are

$$K_0 = 1.0 \text{ m}^2/\text{s}, \quad N_{\text{ref}} = 0.01 \text{ s}^{-1}, \quad S_{\text{ref}} = 0.01 \text{ s}^{-1}, \quad k_N = 2.0, \quad k_S = 2.0, \quad N_{\text{min}} = 10^{-4} \text{ s}^{-1}.$$

These values were selected to yield cross-plume widening on the order of tens of meters over tens of minutes, consistent with early-stage plume dispersion magnitudes, while keeping the sensitivity coefficients order-one so that N and S materially affect outcomes within the sweep ranges.

2.4 Bulk microphysics closure $\mathcal{M}(q_i; \text{RHi})$

A reduced microphysics tendency is applied as a piecewise linear function of RHi:

$$\mathcal{M}(q_i; \text{RHi}) = \begin{cases} -\lambda_{\text{decay}}(\text{RHi}) q_i, & \text{RHi} < 100\%, \\ +\lambda_{\text{grow}}(\text{RHi}) q_i, & \text{RHi} > 100\%, \\ 0, & \text{RHi} = 100\%, \end{cases} \quad (7)$$

with

$$\lambda_{\text{decay}}(\text{RHi}) = \frac{1 - \text{RHi}/100}{\tau_{\text{micro}}}, \quad \lambda_{\text{grow}}(\text{RHi}) = \gamma \frac{\text{RHi}/100 - 1}{\tau_{\text{micro}}}. \quad (8)$$

Here τ_{micro} is a microphysical e-folding timescale and $\gamma \ll 1$ reduces growth strength relative to decay strength in this bulk proxy model. In the implementation, $\tau_{\text{micro}} = 90$ s and $\gamma = 0.05$.

The objective is to represent the correct *direction* of microphysical tendency with minimal complexity:

- For $\text{RHi} < 100\%$, ice sublimates and q_i decays exponentially.
- For $\text{RHi} > 100\%$, supersaturation supports persistence; weak growth prevents the model from forcing immediate decay when dilution is modest.

This closure is designed for regime identification and parameter sensitivity rather than detailed particle-size evolution.

2.5 Domain, initial condition, and boundary conditions

The cross-plume domain has width $L_y = 200$ m and is discretized with $N_y = 201$ points, so $\Delta y = L_y/(N_y - 1)$. The initial plume is Gaussian:

$$q_i(y, 0) = q_{i,0} \exp\left(-\frac{y^2}{2\sigma_y^2}\right), \quad (9)$$

with $q_{i,0} = 10^{-5}$ kg/kg and $\sigma_y = 15$ m.

At the domain boundaries, a no-flux condition is applied:

$$\left. \frac{\partial q_i}{\partial y} \right|_{y=\pm L_y/2} = 0, \quad (10)$$

implemented numerically as a Neumann-type mirrored boundary (ghost-cell) approximation. The finite domain and no-flux boundary prevent artificial mass loss at the edges, under the assumption that the domain is sufficiently wide that the plume remains interior over the simulation window.

2.6 Numerical implementation and stability

Equation (5) is solved using second-order centered finite differences for $\partial^2/\partial y^2$ and forward Euler time integration. For explicit diffusion, the timestep is constrained by the standard stability criterion(14):

$$\Delta t \leq C_{\text{FL}} \frac{\Delta y^2}{K(N, S)}, \quad (11)$$

with a safety factor $C_{\text{FL}} = 0.45$. For each run, Δt is computed from (11), the number of steps is set so that the integration ends exactly at $t_{\text{end}} = 1800$ s, and non-negativity is enforced by clamping $q_i \geq 0$ after each update. This approach ensures stable and consistent integration across large sweeps without manual timestep tuning.

2.7 Diagnostics and persistence definition

Three diagnostic outputs are used throughout the analysis:

Centerline persistence lifetime. Contrail lifetime is defined as the first time the centerline value drops below a fraction α of its initial value:

$$\tau_{\text{persist}} = \min \{t : q_i(0, t) < \alpha q_i(0, 0)\}, \quad (12)$$

with $\alpha = 0.20$. If the criterion is not met before t_{end} , then τ_{persist} is capped at $t_{\text{end}} = 1800$ s.

Total ice proxy. A domain-integrated proxy for ice content is computed as

$$Q_i(t) = \int_{-L_y/2}^{L_y/2} q_i(y, t) dy. \quad (13)$$

This distinguishes microphysical changes (which modify Q_i) from pure redistribution by diffusion (which preserves Q_i in the absence of $\mathcal{M}$).

Effective plume width. A second-moment width is computed as

$$w(t) = \left[\frac{\int (y - \bar{y})^2 q_i(y, t) dy}{\int q_i(y, t) dy} \right]^{1/2}, \quad \bar{y}(t) = \frac{\int y q_i(y, t) dy}{\int q_i(y, t) dy}. \quad (14)$$

This provides a robust measure of spreading and connects to diffusion expectations (e.g. $w^2 \sim w_0^2 + 2Kt$) (12; 14) when microphysical tendencies are weak.

2.8 Sweep design and regime classification

The MATLAB study uses three complementary sweep types:

1. **Coarse sweep table** over $(\text{RHi}, N, S) \in \{90, 100, 110\} \times \{0.005, 0.010\} \times \{0.002, 0.010\}$ to establish baseline sensitivities.
2. **One-dimensional sweeps** varying one control while holding the other two fixed. For N and S sweeps, RHi is set to 95% to ensure microphysical decay is active and dynamical dilution can modulate lifetime.
3. **Two-dimensional regime mapping** in (RHi, N) at fixed $S = 0.005 \text{ s}^{-1}$ to identify persistence boundaries.

A persistence threshold of $\tau_{\text{persist}} \geq 600$ s (10 minutes) is used to classify cases as *persistent* vs. *short-lived* for phase-diagram interpretation. This binary regime definition is convenient for later comparison against ERA5/pycontrails-derived persistence indicators.

2.9 MATLAB results figures

Figures 2–4 summarize the reduced-model behavior. All figures are direct outputs from `contrail_plume_model.m`.

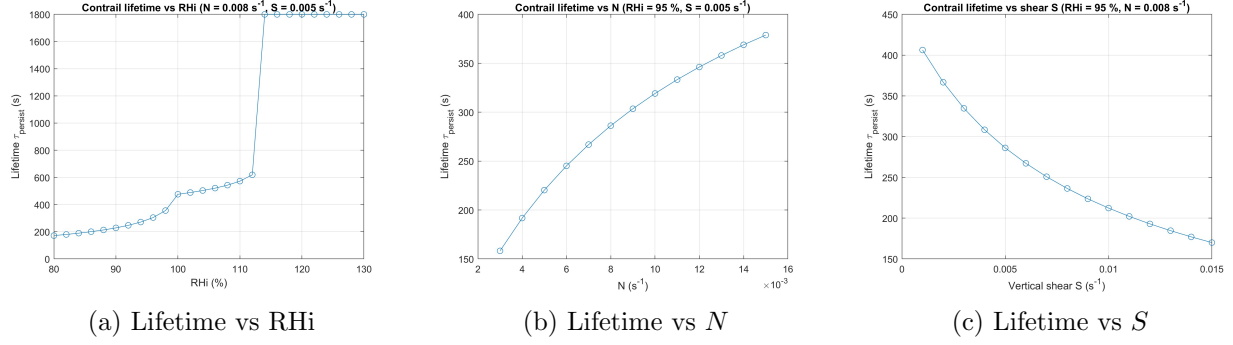


Figure 2: One-dimensional sensitivity sweeps of persistence lifetime τ_{persist} in the MATLAB reduced model. Panel (a) shows a sharp increase of lifetime with ice supersaturation (RH), including cases that reach the simulation cap at $t_{\text{end}} = 1800 \text{ s}$ for sufficiently high RH. Panels (b) and (c) use marginal humidity ($\text{RH} = 95\%$) so that microphysical decay is active and dynamical controls (N and S via $K(N, S)$) measurably modulate lifetime.

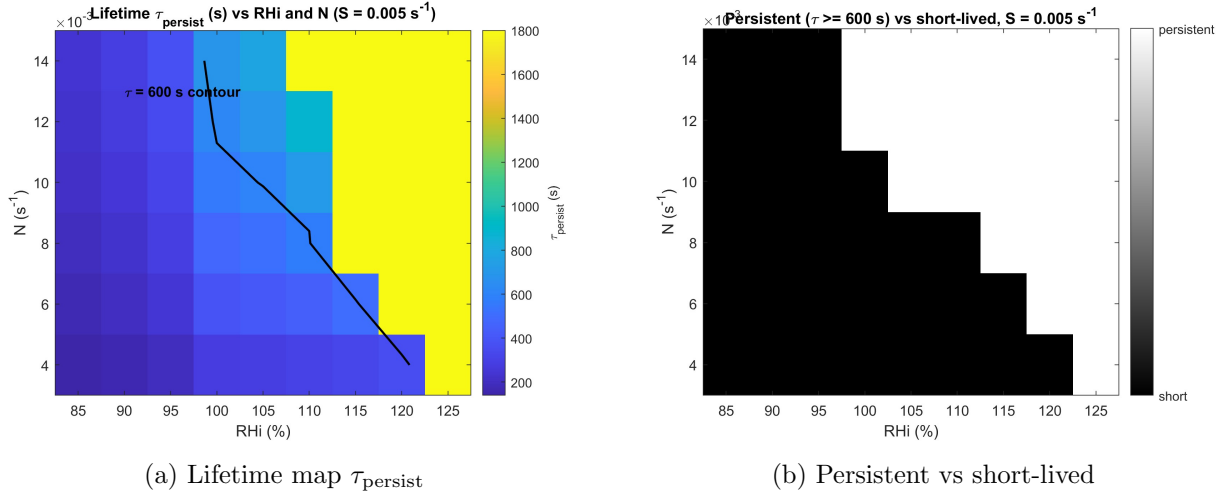
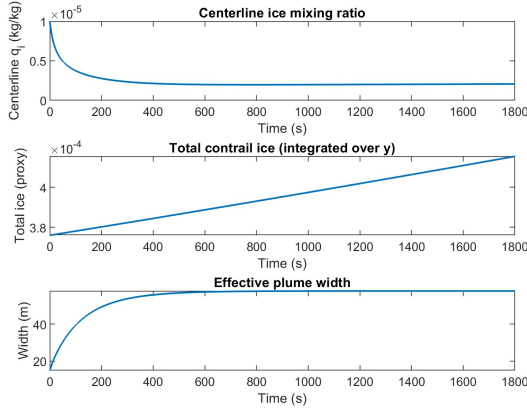


Figure 3: Regime behavior in (RH, N) space at fixed $S = 0.005 \text{ s}^{-1}$. (a) Lifetime phase map with the $\tau_{\text{persist}} = 600 \text{ s}$ contour overlaid. (b) Binary regime classification using the same 600 s persistence threshold. The resulting boundary highlights a nonlinear transition governed primarily by RH and modulated by dynamical dilution through the N -dependence of $K(N, S)$.

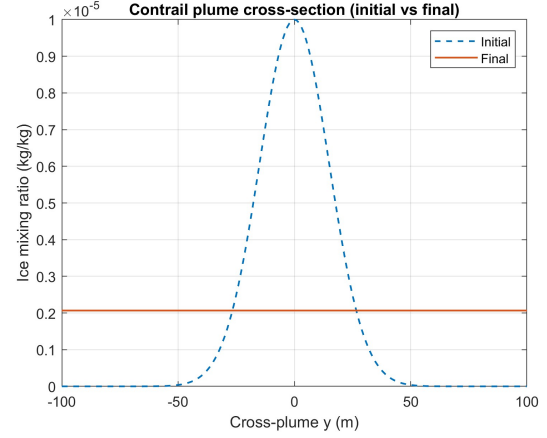
2.10 Recorded MATLAB command-window output

The MATLAB command-window output from the coarse sweep, the detailed demo case, and the regression fit is reproduced below:

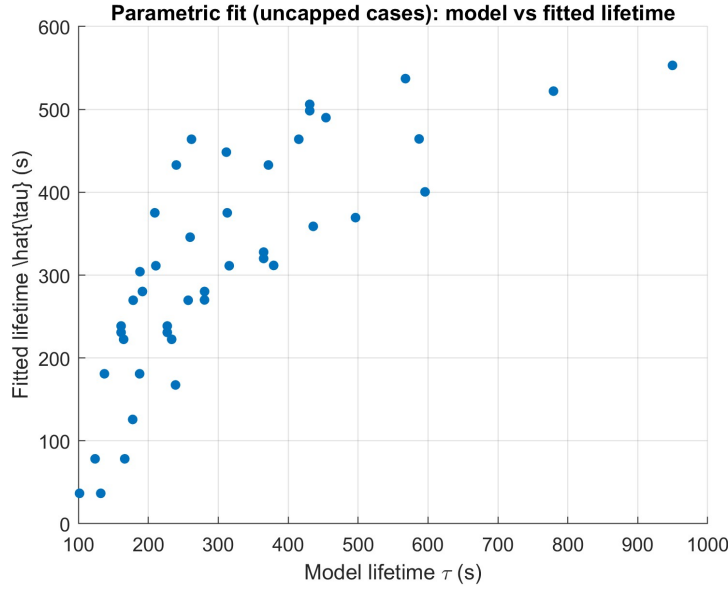
```
=== Basic parameter sweep: lifetime vs (RH, N, S) ===
RH(%)  N (s^-1)  S (s^-1)  Lifetime (s)
  90    0.0050  0.0020    228.4
  90    0.0050  0.0100    136.2
  90    0.0100  0.0020    310.3
  90    0.0100  0.0100    194.1
 100    0.0050  0.0020    475.8
```



(a) Time evolution diagnostics



(b) Initial vs final profile



(c) Parametric fit (uncapped cases)

Figure 4: Mechanistic diagnostics and empirical closure for the MATLAB model. (a) Centerline $q_i(0, t)$, total ice proxy $Q_i(t)$, and plume width $w(t)$ for a representative case $(RH_i, N, S) = (110\%, 0.005 \text{ s}^{-1}, 0.002 \text{ s}^{-1})$. (b) Initial and final cross-plume profiles for the same case, illustrating dilution-driven reduction of the core amplitude, which underlies the persistence definition in (12). (c) Comparison of modeled lifetimes to fitted lifetimes for uncapped cases, supporting a compact empirical lifetime law used as a summary statistic within the explored parameter range.

100	0.0050	0.0100	222.0
100	0.0100	0.0020	793.0
100	0.0100	0.0100	370.1
110	0.0050	0.0020	572.4
110	0.0050	0.0100	237.4
110	0.0100	0.0020	1800.0
110	0.0100	0.0100	420.2


```

=== Detailed demo case ===
RHi = 110 %, N = 0.0050 s-1, S = 0.0020 s-1
Estimated contrail lifetime: 572.4 s

=== Regression fit for tau_persist(RHi, N, S) ===
Fitted coefficients (using 43 / 48 samples):
tau = a0 + a1*(RHi-100)_+ + a2*N-1 + a3*|S|-1
a0 = 300.377 s
a1 = 17.831 s / percent
a2 = -1.346 s2
a3 = 0.801 s2
RMSE of fit (uncapped lifetimes only): 112.92 s

```

2.11 Empirical lifetime law (regression summary)

To provide a compact summary statistic for the reduced-model behavior within the explored parameter range, an empirical law was fit to simulated lifetimes using a simple linear regression on hand-crafted features:

$$\hat{\tau} = a_0 + a_1(\text{RHi} - 100)_+ + a_2 N^{-1} + a_3 |S|^{-1}, \quad (x)_+ \equiv \max(x, 0). \quad (15)$$

The fit was performed on *uncapped* cases only ($\tau_{\text{persist}} < t_{\text{end}}$) to avoid bias from the artificial upper bound at $t_{\text{end}} = 1800$ s. The fitted coefficients and RMSE are listed in Section 2.10. This regression is not asserted as a universal physical law; rather, it offers an interpretable equation for rapid evaluation and for linking the reduced-model response surface to ERA5/*pycontrails*-derived environmental distributions.

3 High-Fidelity Plume Simulations (COMSOL)

The reduced-order MATLAB model presented in Section 2 represents plume spreading through a scalar effective diffusivity $K(N, S)$ and models persistence via a centerline decay criterion. While this approach enables efficient regime mapping, it does not explicitly resolve the spatial structure of the plume, shear-driven deformation, or the coupled velocity–pressure dynamics of the flow. To evaluate the validity of these assumptions and to provide a spatially resolved reference, we perform high-fidelity numerical simulations using COMSOL Multiphysics.

The COMSOL simulations occupy an intermediate tier in a hierarchical modeling framework: they are more physically resolved than the 1D reduced model, but still simplified relative to full atmospheric large-eddy simulations. Their role is to resolve early-time plume transport and dilution mechanisms and to provide diagnostics that can be directly compared to those extracted from the MATLAB model.

3.1 Governing equations

The COMSOL simulations are performed on a two-dimensional Cartesian domain (x, y) . The flow is modeled using the incompressible Navier–Stokes equations,

$$\nabla \cdot \mathbf{u} = 0, \quad (16)$$

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f}, \quad (17)$$

where $\mathbf{u}(x, y, t)$ is the velocity field, $p(x, y, t)$ is pressure, ρ is air density, and μ is the dynamic viscosity. A prescribed forcing $\mathbf{f}$ and/or boundary conditions are used to establish a controlled inflow and vertical shear.

Transport of a plume-linked scalar $c(x, y, t)$ is governed by an advection–diffusion–decay equation,

$$\frac{\partial c}{\partial t} + \mathbf{u} \cdot \nabla c = \nabla \cdot (D \nabla c) - \lambda c, \quad (18)$$

where D is a diffusivity and λ is a bulk decay rate. The scalar c serves as a proxy for exhaust or ice-related quantities and is used to quantify plume dilution and persistence. Equation (18) is the spatially resolved analogue of the reduced-order evolution equation (??).

3.2 Domain, initial conditions, and boundary conditions

The computational domain extends approximately $L_x \sim 200$ m in the streamwise direction and $L_y \sim 200$ m in the cross-stream direction. A localized plume is introduced near the inlet at $t = 0$, representing the injection of exhaust into an ambient background flow.

The initial scalar distribution is compact and centered on the plume axis, with a Gaussian cross-stream structure consistent with the reduced-order model,

$$c(x, y, 0) \propto \exp\left(-\frac{y^2}{2\sigma^2}\right), \quad (19)$$

where σ sets the initial plume width. Temperature and velocity fields are initialized with localized perturbations near the plume source.

Boundary conditions are specified to minimize artificial reflection and mass loss:

- **Inlet:** prescribed inflow velocity and scalar concentration profile.
- **Outlet:** open (convective) boundary allowing momentum and scalar to exit.
- **Upper and lower boundaries:** slip or weakly constrained velocity conditions and zero-flux (Neumann) conditions for the scalar.

3.3 Imposed shear and relation to reduced-order parameters

Vertical shear is imposed through the velocity field such that the streamwise velocity varies with the cross-stream coordinate y . The characteristic shear magnitude is defined as

$$S \equiv \left| \frac{\partial u_x}{\partial y} \right|, \quad (20)$$

which directly corresponds to the shear parameter S used in the reduced-order MATLAB model. By explicitly resolving shear-driven deformation of the plume, the COMSOL simulations enable assessment of whether shear effects can be adequately summarized by a scalar effective diffusivity.

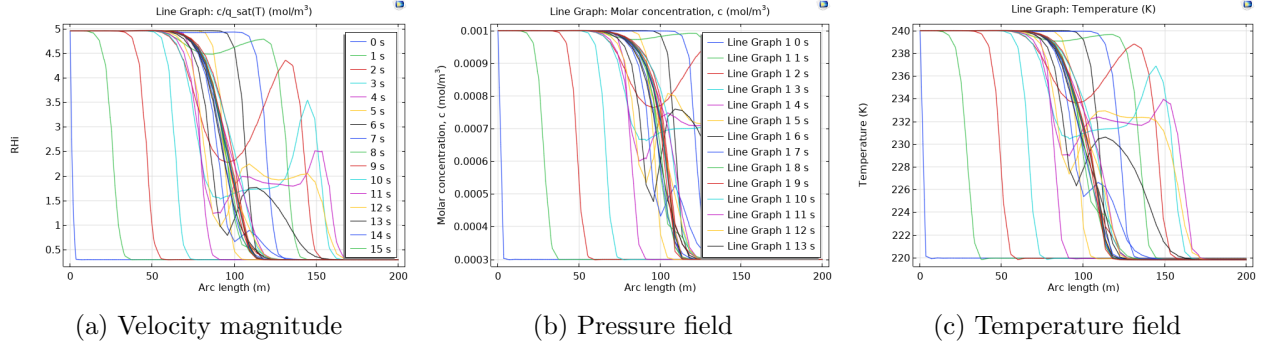


Figure 5: Resolved COMSOL fields at initialization ($t = 0$). The injected plume appears as a localized region of elevated velocity and temperature near the inlet, while the surrounding flow remains quiescent.

3.4 Resolved initial fields

Figure 5 shows the resolved COMSOL fields at initialization ($t = 0$). A localized region of elevated velocity magnitude and temperature marks the initial plume near the inlet. Pressure gradients are strongest near the source region, reflecting the coupled nature of the velocity–pressure solution.

3.5 Time evolution and animated diagnostics

The temporal evolution of the plume is visualized using animated COMSOL exports. Because animated GIFs cannot be embedded directly in the paper, they are provided as external links and referenced here:

Flow and thermodynamic evolution (click on links to see very cool and hard worked on GIFs :D)

- Velocity field evolution: [GIF of Velocity field](#)
- Pressure evolution: [GIF of Pressure](#)
- Temperature evolution: [GIF of Temperature](#)

Moisture and scalar dilution

- Relative humidity over ice (RH_i) evolution: [GIF of RH_i](#)
- Scalar concentration evolution: [GIF of Scalar Concentration](#)
- Combined diagnostic animation: [GIF of combined animation](#)

These animations show the plume advecting downstream while spreading laterally under the combined influence of shear and diffusion. The scalar concentration remains confined to the plume core at early times and progressively decays as dilution dominates.

3.6 Diagnostic extraction and comparison to the reduced model

To enable quantitative comparison with the reduced-order model, COMSOL outputs are post-processed to extract diagnostics analogous to those defined in Section 2.7. In particular, cross-stream profiles of $c(y, t)$ at fixed downstream locations are used to compute an effective plume width,

$$w(t) = \left[\frac{\int (y - \bar{y})^2 c(y, t) dy}{\int c(y, t) dy} \right]^{1/2}, \quad (21)$$

from which an effective diffusivity may be inferred using the diffusion scaling

$$K_{\text{eff}}(t) = \frac{w^2(t) - w_0^2}{2t}. \quad (22)$$

Centerline decay of the scalar concentration provides an additional persistence proxy, which can be compared directly to the centerline-based lifetime definition used in the MATLAB model. The agreement between COMSOL-derived K_{eff} and the parameterized $K(N, S)$ supports the effective-diffusivity abstraction. deviations indicate regimes where shear-driven deformation or non-diffusive transport becomes important.

I, unfortunately, misplaced the Matlab file and could not find the exports in the hundreds of plots generated for this project and hence it has not been shown here.

4 Reanalysis-Driven Environmental Diagnostics (ERA5/pycontrails)

The MATLAB reduced-order model and the COMSOL plume simulations quantify how contrail persistence depends on three environmental controls, ice supersaturation (RHi), static stability (N), and vertical shear (S), when these parameters are treated as tunable inputs. To anchor those parameter regimes in realistic atmospheric variability, ERA5 reanalysis data were processed using the `pycontrails` workflow to compute upper-tropospheric fields relevant to contrail formation and persistence.

4.1 Thermodynamic feasibility: ice supersaturation in space, time, and along-track

Persistent contrails require ambient air to be supersaturated with respect to ice ($\text{RHi} > 100\%$), so the first question is whether the upper-tropospheric environment contains sustained ice-supersaturated regions (ISSRs) in the relevant pressure range. Figure 6 shows a representative RHi field at 250 hPa (00Z on 2024-10-10) together with the sampled RHi along a cruise-level trajectory through western/central Europe. The map reveals that ISSRs are not uniformly distributed: they occur as narrow, filamentary structures embedded within much larger sub-saturated areas. This spatial intermittency is consistent with the conceptual picture that contrail persistence is conditional on passing through specific meteorological features rather than being guaranteed at cruise altitude.

The along-track RHi time series further illustrates that the aircraft encounters long stretches of sub-saturated air punctuated by a limited supersaturated segment. In the present trajectory, only a small fraction of waypoints exceed 100% RHi (approximately 4%), implying that even before dynamical dilution is considered, persistent-contrail conditions are relatively rare along the track. This fact provides a direct interpretive link to the MATLAB phase behavior in Fig. 2–3: the sharp transition in persistence near $\text{RHi} \approx 100\%$ corresponds, in the reanalysis, to a sparse set of spatiotemporal organized events.

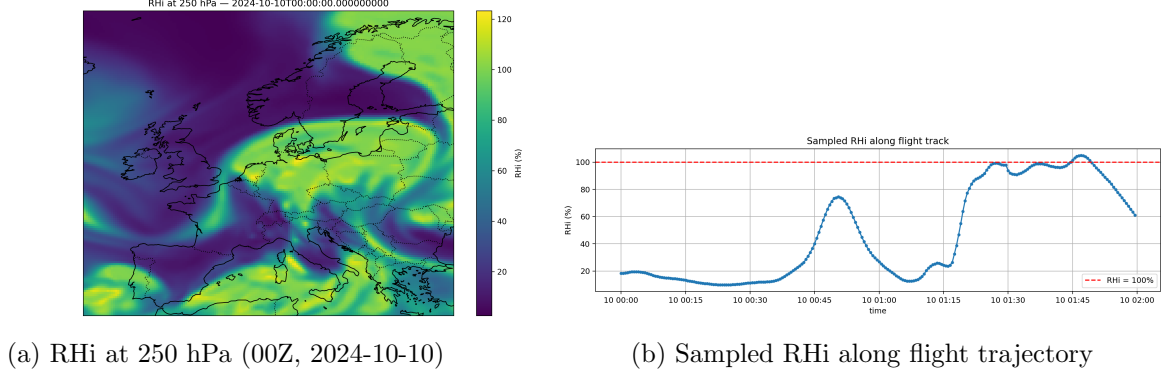


Figure 6: Ice supersaturation in the upper troposphere from ERA5/pycontrails. (a) A representative RHi field at 250 hPa exhibits spatially intermittent ISSRs ($\text{RHi} > 100\%$). (b) Along-track sampling shows extended sub-saturated segments and a limited supersaturated interval, motivating a persistence framework that is conditional on thermodynamic feasibility.

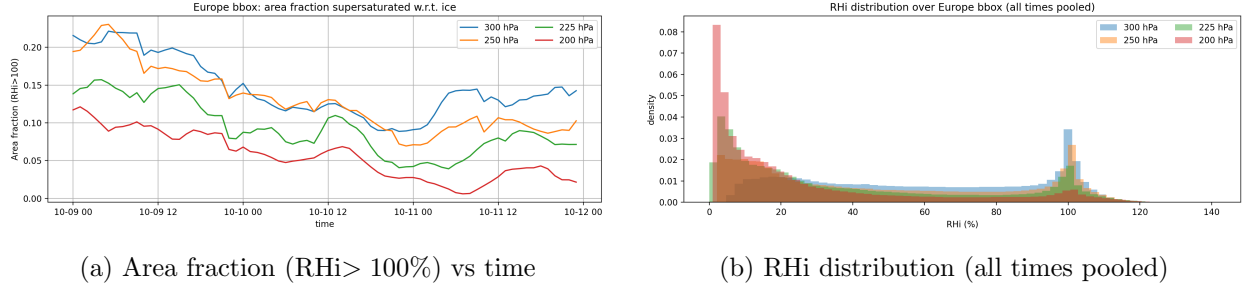


Figure 7: Regional context for ice supersaturation over Europe. (a) The fraction of the domain that is ice-supersaturated varies with synoptic evolution and altitude, indicating that persistence potential is meteorology-dependent. (b) Pooled histograms show that supersaturation occurs in the tail of a skewed distribution, consistent with the limited along-track supersaturation fraction.

To contextualize the single-track sampling, Europe-wide statistics were computed over the full domain and all available analysis times. Figure 7 summarizes the areal prevalence of ISSRs and the distribution of RHi at multiple pressure levels. The area fraction with $\text{RHi} > 100\%$ varies systematically with both time and altitude, typically spanning $\mathcal{O}(5\text{--}20\%)$ depending on level. Higher levels (e.g. 300–250 hPa) generally show larger supersaturated fractions than lower levels (225–200 hPa), consistent with colder ambient temperatures and upper-level dynamical forcing. The pooled histograms display a strongly skewed RHi distribution with a long tail beyond saturation, reinforcing that ISSRs represent the high-humidity tail of upper-tropospheric variability rather than the mean state.

4.2 Dynamical controls: stability and shear as reanalysis analogues of plume transport

While ISSRs determine whether ice growth is thermodynamically possible, the persistence lifetime and morphology depend on the rate at which the plume mixes and deforms. In the MATLAB model, these effects were encoded through an effective diffusivity $K(N, S)$ (Eq. (6)), while the COMSOL simulations resolved advection–diffusion and shear-driven deformation explicitly. ERA5

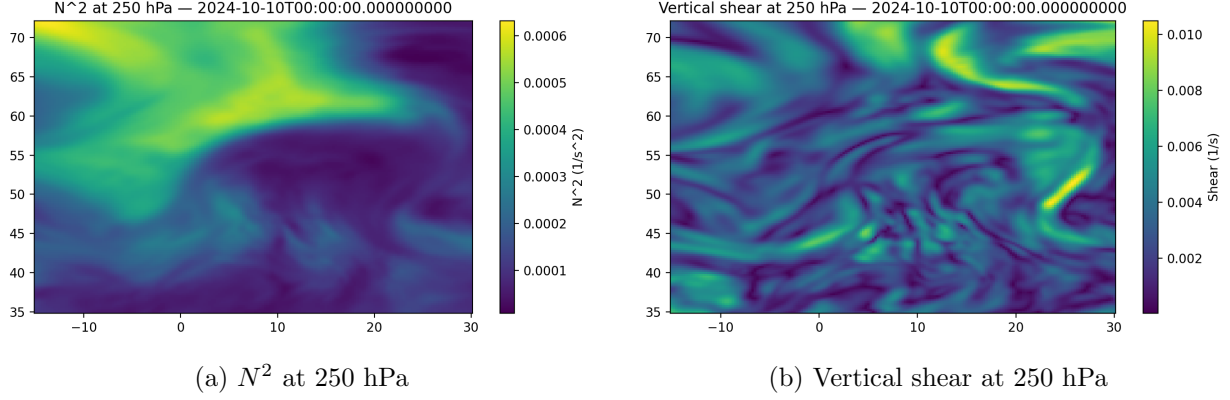


Figure 8: Dynamical controls on contrail persistence from ERA5. Static stability (N^2) varies smoothly on synoptic scales, while vertical shear exhibits sharper gradients associated with strong-flow features. These diagnostics provide meteorologically grounded analogues for the transport/dilution controls represented by $K(N, S)$ in the MATLAB model and resolved explicitly in COMSOL.

provides the corresponding dynamical controls via stability and vertical shear diagnostics.

Potential temperature was computed as $\theta = T(p_0/p)^{R/c_p}$, and the Brunt–Väisälä frequency squared was obtained from (16)

$$N^2 = \frac{g}{\theta} \frac{\partial \theta}{\partial z}. \quad (23)$$

using finite differences in the vertical after converting geopotential to geometric height $z_{\text{geo}} = z/g$. Vertical shear was computed from (16)

$$S = \sqrt{\left(\frac{\partial u}{\partial z}\right)^2 + \left(\frac{\partial v}{\partial z}\right)^2}. \quad (24)$$

Figures 8 show representative maps of N^2 and S at 250 hPa. The N^2 field exhibits smooth, synoptic-scale structure consistent with large-scale stratification, while the shear field shows sharper filaments associated with jets and frontal zones. The magnitudes obtained are physically reasonable for the upper troposphere (here, $N^2 \sim 10^{-5}$ – 10^{-4} s^{-2} and $S \sim 10^{-3}$ – 10^{-2} s^{-1}), placing the reanalysis environment squarely within the sweep ranges explored in MATLAB and the deformation regimes represented in COMSOL.

4.3 A unified persistence indicator and connection to plume-scale results

To connect the reanalysis environment to the reduced-order persistence picture, a compact persistence indicator was constructed along the trajectory using three ingredients: (i) humidity availability (RH_i), (ii) stratification (via N^2), and (iii) deformation/mixing potential (via shear S). A “raw” score was first defined as a weighted sum of normalized components,

$$\mathcal{P}_{\text{raw}} = w_{\text{RH}_i} \widetilde{\text{RH}_i} + w_N \widetilde{N^2} + w_S \left(1 - \widetilde{S}\right), \quad w_{\text{RH}_i} + w_N + w_S = 1, \quad (25)$$

where tildes denote robust normalization to $[0, 1]$ using percentile scaling. The factor $1 - \widetilde{S}$ encodes the expectation that stronger shear accelerates plume deformation and dilution, reducing persistence of a coherent core. This representation mirrors the MATLAB closure (Eq. (6)) in which

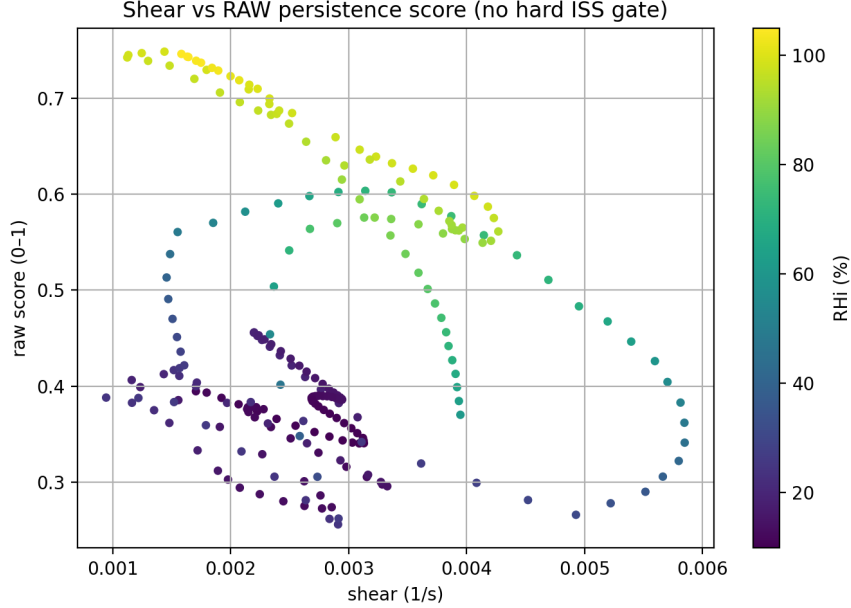


Figure 9: Relationship between vertical shear and a raw persistence favorability score along the flight, colored by RH_i. Higher shear is associated with lower persistence favorability, while higher humidity shifts the score upward across the shear range. This mirrors the MATLAB sensitivity of persistence to shear under marginal humidity and is consistent with shear-driven deformation/dilution mechanisms resolved in COMSOL.

increasing shear increases effective mixing, and it is consistent with COMSOL behavior in which shear-driven stretching enhances dilution rates along the cross-plume direction.

Figure 9 shows the relationship between vertical shear and $\mathcal{P}_{\text{raw}}$, colored by RH_i. Two features are prominent. First, there is a clear anticorrelation between shear and the raw score: high shear environments systematically correspond to lower persistence favorability. Second, humidity modulates this relationship: high-RH_i points populate the upper envelope of $\mathcal{P}_{\text{raw}}$, while low-RH_i points cluster at lower scores. This joint dependence provides a reanalysis analogue of the MATLAB regime behavior: the MATLAB sweeps show that near-threshold humidity is particularly sensitive to dynamical dilution, and the reanalysis scatter exhibits that the highest persistence favorability occurs when both humidity and dynamical conditions align.

Because persistent contrails require (at minimum) near-ice-saturated conditions, a moisture gating factor was introduced to softly enforce thermodynamic feasibility while avoiding an overly brittle binary threshold. The gating function was chosen as a sigmoid centered at RH_i= 100%,

$$\mathcal{G}(\text{RH}_i) = (1 + \exp[-k(\text{RH}_i - 100)])^{-1}, \quad (26)$$

with k controlling the sharpness of transition. The final soft-gated indicator is $\mathcal{P}_{\text{soft}} = \mathcal{P}_{\text{raw}}\mathcal{G}(\text{RH}_i)$. Figure 10 shows the gating curve and the resulting shear–score relationship. As expected, the soft-gated score collapses toward zero in strongly sub-saturated air and highlights a small subset of waypoints where persistence is plausible. In the present trajectory, only $\approx 4\%$ of waypoints are ice-supersaturated, which explains the sparse set of elevated $\mathcal{P}_{\text{soft}}$ values. This sparsity is not a numerical artifact. It is a direct expression of the thermodynamic threshold that also appears in the MATLAB lifetime sweeps.

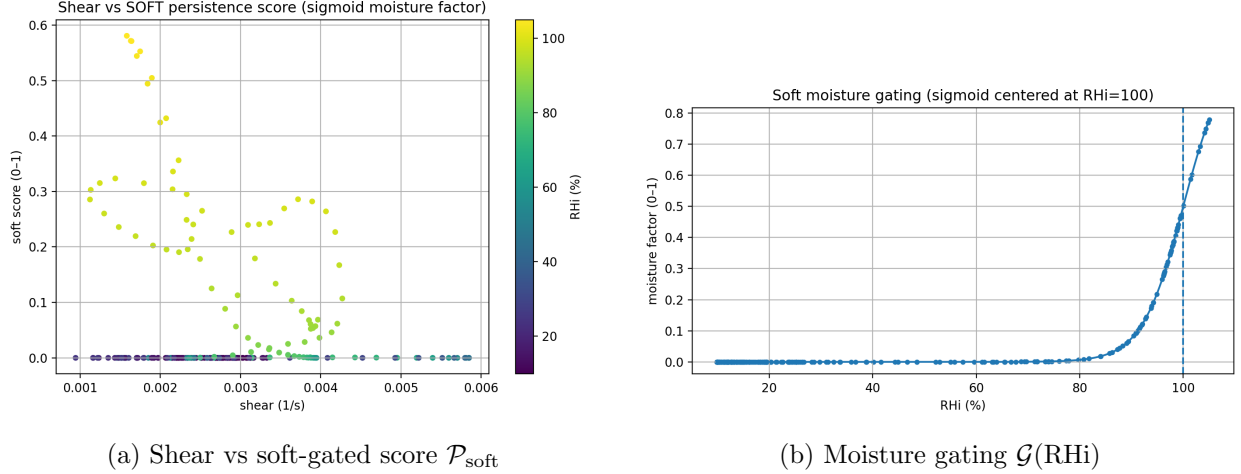


Figure 10: Soft enforcement of thermodynamic feasibility for persistence. A sigmoid gating centered at $\text{RH}_i = 100\%$ suppresses persistence in strongly sub-saturated air while allowing continuous transitions near the threshold. The resulting $\mathcal{P}_{\text{soft}}$ highlights a small set of waypoints as plausibly persistent, consistent with the limited ISS prevalence in both the along-track sampling and the regional statistics.

4.4 Synthesis across scales: interpreting ERA5 diagnostics through MATLAB and COMSOL

The combined ERA5/pycontrails diagnostics provide a consistent bridge between the idealized parameter sweeps (MATLAB) and the resolved transport physics (COMSOL). The MATLAB model predicts a sharp persistence transition near ice saturation and systematic modulation by shear and stability when humidity is marginal. The reanalysis results show exactly the environmental structure required for that picture: ISSRs occupy a small, organized portion of the domain (and a small fraction of the flight), while shear and stability vary across the same order-of-magnitude ranges used in the reduced model. COMSOL, in turn, supplies mechanistic support for interpreting shear as a deformation-driven dilution pathway, which rationalizes the anticorrelation between shear and persistence favorability seen in the raw-score scatter.

Overall, the multiscale results support a two-part persistence criterion: contrails are thermodynamically feasible only within localized ISSRs, and their subsequent persistence is regulated by the competition between stabilizing stratification and shear-driven deformation and mixing. This framing provides a physically grounded pathway for using reanalysis meteorology to interpret and constrain plume-scale persistence models without relying on satellite detection or case-by-case manual classification.

5 Summary and Conclusions

5.1 Hypothesis check

This project began with the hypothesis that contrail persistence is primarily gated by **ice supersaturation** (RH_i), with **dynamical dilution** set by stability (N) and vertical shear (S) modulating lifetime once conditions are near threshold. Across the three modeling tiers developed here, the results support that hypothesis:

- **Thermodynamic control (RHi) is the dominant gate.** The MATLAB reduced-order model shows a sharp transition in the persistence proxy τ_{persist} near $\text{RHi} \approx 100\%$, consistent with the physical requirement that ice cannot be maintained in subsaturated air. The ERA5/pycontrails diagnostics show that ice-supersaturated regions (ISSRs) are spatially intermittent and occupy a limited fraction of the domain/trajectory, meaning that persistence-capable conditions are rare in both space and time.
- **Shear and stability modulate persistence under marginal humidity.** When RHi is near (but below) saturation, the MATLAB sweeps demonstrate systematic sensitivity of lifetime to the effective mixing control $K(N, S)$: stronger shear shortens lifetime via enhanced dilution, while stability changes mixing rates and therefore the decay of the plume core. In the reanalysis-based indicator, higher shear is associated with lower persistence favorability, consistent with the same mechanism.
- **Mechanistic interpretation is consistent across tiers.** COMSOL resolves the spatial plume deformation and advection–diffusion physics that are parameterized in the reduced model. The resulting picture is coherent: *RHi determines whether persistence is feasible at all, and transport controls determine how rapidly the plume core dilutes once feasible.*

5.2 What we learned about the question

The central question was: *How do RHi, static stability, and vertical wind shear jointly control contrail plume persistence?* The main takeaways are:

1. **Persistence is a two-part condition: feasibility + survivability.** Feasibility is set by the thermodynamic threshold (RHi near or above 100%), while survivability depends on how quickly mixing reduces centerline concentration/ice proxy below a detectability or “persistence” threshold.
2. **Near-threshold regimes are the most sensitive.** The strongest dynamical sensitivity occurs when RHi is close to saturation, where modest changes in mixing rates can shift the system between short-lived and persistent behavior (seen clearly in MATLAB phase behavior and in the reanalysis scatter/soft-gated score).
3. **A practical interpretation for meteorology-driven prediction emerges.** In reanalysis space, ISSRs identify where persistence is possible. Then shear/stability diagnostics provide a physically interpretable ordering of “more” vs “less” persistence-favorable environments.

5.3 Limitations

The framework is intentionally hierarchical and therefore simplified: (i) the MATLAB microphysics is bulk/parametric rather than particle-resolved, (ii) COMSOL is 2D and does not represent full 3D turbulence/microphysics coupling, and (iii) ERA5 provides resolved-scale meteorology but not subgrid supersaturation variability.

6 Future Work

Several follow-up directions would materially strengthen the answer to the persistence question and move the framework closer to predictive capability:

1. **Upgrade the microphysics in the reduced model.** Replace the bulk piecewise growth/decay closure with a minimal but physical scheme (e.g., ice crystal number/mass evolution with temperature-dependent saturation vapor pressure, deposition/sublimation kinetics, and sedimentation). This would allow the RHi threshold behavior to emerge from physics rather than being imposed.
2. **Calibrate $K(N, S)$ using COMSOL-inferred transport metrics.** Use COMSOL time-dependent plume widths $w(t)$ to infer $K_{\text{eff}}(t)$ and fit a stability/shear closure that better represents shear-driven stretching versus purely diffusive spreading. This would tighten the link between the reduced model and resolved dynamics.
3. **Add radiative relevance metrics.** Extend “persistence” beyond a centerline threshold by coupling to simple optical depth or ice water path proxies, enabling a clearer connection to radiative forcing rather than lifetime alone.
4. **Expand the meteorological sampling.** Repeat the ERA5/pycontrails workflow across many days/seasons and multiple flight tracks to quantify how often the atmosphere occupies different regions of the (RHi, N, S) phase space, and to identify regimes associated with jets/frontal zones versus quiescent conditions.
5. **Validation against observations and/or higher-fidelity models.** Compare predicted persistence regimes against satellite-based contrail datasets or case-study observations, and (if available) benchmark against LES contrail studies to quantify biases from 2D transport assumptions and simplified microphysics.

7 Code Availability

All custom code used in this study is documented below to support transparency and reproducibility.

7.1 Python (ERA5 and pycontrails)

All Python-based reanalysis processing, trajectory sampling, and diagnostic plotting was performed in a Google Colab notebook, which includes all dependencies, data access instructions, and executable cells.

Google Colab notebook: <https://colab.research.google.com/drive/1f880jXfbD1PuoSE94C86UB5B2F600TscrollTo=bTiiFLwMhFk9&uniqifi>

7.2 MATLAB reduced-order plume model

The reduced-order plume model described in Section 2 was implemented entirely in MATLAB. The full source code used to generate all MATLAB figures and results is provided below.

```
% =====
% contrail_plume_model.m
% Reduced-order 1D contrail plume model

clear; close all; clc;
```

```

% Example environmental values for a basic sweep (coarse table)
RHi_vals_basic = [90, 100, 110];      % Relative humidity w.r.t. ice (%)
N_vals_basic   = [0.005, 0.010];      % Brunt{Väisälä frequency ( $s^{-1}$ )
S_vals_basic   = [0.002, 0.010];      % Vertical wind shear ( $s^{-1}$ )

% Numerical / physical parameters grouped in a struct
params = struct();
params.Ly      = 200;      % domain width in meters (cross-plume)
params.Ny      = 201;      % number of grid points
params.t_end    = 1800;    % max simulation time [s]
params.CFL      = 0.45;    % CFL safety factor for diffusion

params.qi0_center = 1e-5;    % initial centerline ice mixing ratio [kg/kg]
params.plume_sigma_y = 15;    % initial plume half-width [m]

params.K0       = 1.0;      % base diffusivity [ $m^2/s$ ]
params.N_ref    = 0.01;    % reference N for scaling [ $s^{-1}$ ]
params.S_ref    = 0.01;    % reference shear for scaling [ $s^{-1}$ ]
params.kN       = 2.0;      % sensitivity of K to stability
params.kS       = 2.0;      % sensitivity of K to shear

% Microphysics: decay if RHi < 100%, weak growth if RHi > 100%
params.tau_micro = 90;      % microphysical e-folding time [s]
params.threshold_frac = 0.20; % lifetime: centerline < 20% of initial

params.T_env    = 220;      % K (not used directly here)

% Lifetime threshold for "persistent" classification (e.g. 10 minutes)
tau_persist_thresh = 600;    % [s]

fprintf('=== Basic parameter sweep: lifetime vs (RHi, N, S) ===\n');
fprintf('RHi(%%)\tN ( $s^{-1}$ )\tS ( $s^{-1}$ )\tLifetime (s)\n');

for iR = 1:numel(RHi_vals_basic)
    for iN = 1:numel(N_vals_basic)
        for iS = 1:numel(S_vals_basic)
            RHi = RHi_vals_basic(iR);
            N    = N_vals_basic(iN);
            S    = S_vals_basic(iS);

            result = simulate_contrail_1D(RHi, N, S, params);
            tau_persist = result.tau_persist;

            fprintf('%5.0f\t%7.4f\t%7.4f\t%8.1f\n', RHi, N, S, tau_persist);
        end
    end
end
end

```

```

% ----- Sweep in RHi (holding N, S fixed) -----
RHi_vec = 80:2:130;
N_fixed_for_RHi = 0.008;
S_fixed_for_RHi = 0.005;

tau_RHi = zeros(size(RHi_vec));

for i = 1:numel(RHi_vec)
    result = simulate_contrail_1D(RHi_vec(i), N_fixed_for_RHi, S_fixed_for_RHi, params);
    tau_RHi(i) = result.tau_persist;
end

figure;
plot(RHi_vec, tau_RHi, 'o-');
xlabel('RHi (%)');
ylabel('Lifetime \tau_{persist} (s)');
title(sprintf('Contrail lifetime vs RHi (N = %.3f s^{-1}, S = %.3f s^{-1})', ...
    N_fixed_for_RHi, S_fixed_for_RHi));
grid on;

% ----- Sweep in N (holding RHi, S fixed) -----
% Use a *marginal* humidity so N actually matters (not fully persistent)
N_vec = 0.003:0.001:0.015;
RHi_fixed_for_N = 95;      % < 100%, so some cases decay
S_fixed_for_N = 0.005;

tau_N = zeros(size(N_vec));

for i = 1:numel(N_vec)
    result = simulate_contrail_1D(RHi_fixed_for_N, N_vec(i), S_fixed_for_N, params);
    tau_N(i) = result.tau_persist;
end

figure;
plot(N_vec, tau_N, 'o-');
xlabel('N (s^{-1})');
ylabel('Lifetime \tau_{persist} (s)');
title(sprintf('Contrail lifetime vs N (RHi = %.0f %, S = %.3f s^{-1})', ...
    RHi_fixed_for_N, S_fixed_for_N));
grid on;

% ----- Sweep in S (holding RHi, N fixed) -----
S_vec = 0.001:0.001:0.015;
RHi_fixed_for_S = 95;      % again use marginal humidity
N_fixed_for_S = 0.008;

```

```

tau_S = zeros(size(S_vec));

for i = 1:numel(S_vec)
    result = simulate_contrail_1D(RHi_fixed_for_S, N_fixed_for_S, S_vec(i), params);
    tau_S(i) = result.tau_persist;
end

figure;
plot(S_vec, tau_S, 'o-');
xlabel('Vertical shear S (s^{-1})');
ylabel('Lifetime \tau_{persist} (s)');
title(sprintf('Contrail lifetime vs shear S (RHi = %.0f %, N = %.3f s^{-1})', ...
    RHi_fixed_for_S, N_fixed_for_S));
grid on;

RHi_vec_2D = 85:5:125;
N_vec_2D    = 0.004:0.002:0.014;
S_fixed_2D = 0.005;

[RR, NN] = meshgrid(RHi_vec_2D, N_vec_2D);
tau_map = zeros(size(RR));

for iN = 1:numel(N_vec_2D)
    for iR = 1:numel(RHi_vec_2D)
        RHi = RHi_vec_2D(iR);
        N    = N_vec_2D(iN);
        result = simulate_contrail_1D(RHi, N, S_fixed_2D, params);
        tau_map(iN, iR) = result.tau_persist;
    end
end

figure;
imagesc(RHi_vec_2D, N_vec_2D, tau_map);
set(gca, 'YDir', 'normal');
xlabel('RHi (%)');
ylabel('N (s^{-1})');
title(sprintf('Lifetime \tau_{persist} (s) vs RHi and N (S = %.3f s^{-1})', ...
    S_fixed_2D));
hcb = colorbar;
ylabel(hcb, '\tau_{persist} (s)');

hold on;
% Only contour if there is some variation
if max(tau_map(:)) > min(tau_map(:))
    contour(RR, NN, tau_map, [tau_persist_thresh tau_persist_thresh], ...
        'k', 'LineWidth', 1.5);
    text(RHi_vec_2D(2), N_vec_2D(end)-0.001, ...

```

```

        sprintf('\tau = %d s contour', tau_persist_thresh), ...
        'Color', 'k', 'FontWeight', 'bold');
else
    warning('tau_map is (almost) constant; skipping contour line.');
```

```

end

persistent_mask = tau_map >= tau_persist_thresh;
figure;
imagesc(RHi_vec_2D, N_vec_2D, persistent_mask);
set(gca, 'YDir', 'normal');
xlabel('RHi (%)');
ylabel('N (s^{-1})');
title(sprintf('Persistent (\tau >= %d s) vs short-lived, S = %.3f s^{-1}', ...
    tau_persist_thresh, S_fixed_2D));
colormap(gray);
colorbar('Ticks',[0 1], 'TickLabels',{'short','persistent'});

RHi_demo = 110;           % highly supersaturated case
N_demo    = 0.005;
S_demo    = 0.002;

[result_demo, grid_demo] = simulate_contrail_1D(RHi_demo, N_demo, S_demo, params, true);

fprintf('\n=== Detailed demo case ===\n');
fprintf('RHi = %.0f %, N = %.4f s^{-1}, S = %.4f s^{-1}\n', RHi_demo, N_demo, S_demo);
fprintf('Estimated contrail lifetime: %.1f s\n', result_demo.tau_persist);

figure;
subplot(3,1,1);
plot(result_demo.time, result_demo.qi_center, 'LineWidth', 1.2);
xlabel('Time (s)');
ylabel('Centerline q_i (kg/kg)');
title('Centerline ice mixing ratio');

subplot(3,1,2);
plot(result_demo.time, result_demo.qi_total, 'LineWidth', 1.2);
xlabel('Time (s)');
ylabel('Total ice (proxy)');
title('Total contrail ice (integrated over y)');

subplot(3,1,3);
plot(result_demo.time, result_demo.width, 'LineWidth', 1.2);
xlabel('Time (s)');
ylabel('Width (m)');
title('Effective plume width');
```

```

figure;
plot(grid_demo.y, result_demo.qi_initial, '--', 'LineWidth', 1.2); hold on;
plot(grid_demo.y, result_demo.qi_final, '--', 'LineWidth', 1.2);
xlabel('Cross-plume y (m)');
ylabel('Ice mixing ratio (kg/kg)');
legend('Initial', 'Final');
title('Contrail plume cross-section (initial vs final)');
grid on;

```

```
do_regression_fit = true;
```

```

if do_regression_fit
    fprintf('\n=== Regression fit for tau_persist(RHi, N, S) ===\n');

    % Coarse grid of samples over a realistic parameter range
    RHi_vec_reg = 85:10:120;
    N_vec_reg   = 0.004:0.003:0.013;
    S_vec_reg   = 0.003:0.004:0.011;

    [RR_reg, NN_reg, SS_reg] = ndgrid(RHi_vec_reg, N_vec_reg, S_vec_reg);
    nSamples = numel(RR_reg);

    X = zeros(nSamples, 3);    % columns: RHi, N, S
    y = zeros(nSamples, 1);    % lifetime

    for i = 1:nSamples
        RHi = RR_reg(i);
        N   = NN_reg(i);
        S   = SS_reg(i);

        result = simulate_contrail_1D(RHi, N, S, params);
        X(i,:) = [RHi, N, S];
        y(i)   = result.tau_persist;
    end

    % Build regression features
    RHi_eff = max(X(:,1) - 100, 0);
    N_inv   = 1 ./ X(:,2);
    S_inv   = 1 ./ abs(X(:,3));

    A = [ones(nSamples,1), RHi_eff, N_inv, S_inv];

    % ---- NEW: fit only on cases that did NOT hit the lifetime cap ----
    cap = params.t_end;
    fit_mask = y < cap - 1e-6;          % "true" lifetimes
    A_fit = A(fit_mask,:);

```

```

y_fit = y(fit_mask);

coeffs = A_fit \ y_fit;

a0 = coeffs(1);
a1 = coeffs(2);
a2 = coeffs(3);
a3 = coeffs(4);

fprintf('Fitted coefficients (using %d / %d samples):\n', ...
        sum(fit_mask), nSamples);
fprintf('tau = a0 + a1*(RHi-100)_+ + a2*N^{-1} + a3*|S|^{-1}\n');
fprintf('a0 = %.3f s\n', a0);
fprintf('a1 = %.3f s / percent\n', a1);
fprintf('a2 = %.3f s^2\n', a2);
fprintf('a3 = %.3f s^2\n', a3);

tau_pred_fit = A_fit * coeffs;
rmse = sqrt(mean((tau_pred_fit - y_fit).^2));
fprintf('RMSE of fit (uncapped lifetimes only): %.2f s\n', rmse);

% Plot model vs fitted for uncapped cases
figure;
scatter(y_fit, tau_pred_fit, 25, 'filled');
xlabel('Model lifetime \tau (s)');
ylabel('Fitted lifetime \hat{\tau} (s)');
title('Parametric fit (uncapped cases): model vs fitted lifetime');
grid on;
end

function [result, grid] = simulate_contrail_1D(RHi, N, S, params, doOutput)
%SIMULATE_CONTRAIL_1D
% Simple 1D diffusion + bulk microphysics model for a contrail cross-section.
%
% Lifetime is defined as the time when the CENTERLINE ice mixing ratio
% drops below threshold_frac * (initial centerline value).
%
% Inputs:
%   RHi      - Relative humidity w.r.t. ice in percent
%   N        - Brunt{Väisälä frequency (s^{-1})
%   S        - Vertical shear magnitude (s^{-1})
%   params   - struct of physical/numerical parameters
%   doOutput - (optional) if true, returns time series for plotting
%
% Outputs:
%   result.tau_persist - estimated lifetime [s]

```



```

%      result.qi_final      - final ice profile [kg/kg]
%      result.qi_initial    - initial ice profile [kg/kg]
%      result.qi_center     - centerline qi(t)
%      result.qi_total      - total ice vs time
%      result.width         - plume width vs time
%      result.time          - time vector
%      grid.y               - cross-plume coordinate [m]

if nargin < 5
    doOutput = false;
end

% Unpack parameters
Ly          = params.Ly;
Ny          = params.Ny;
t_end       = params.t_end;
CFL         = params.CFL;
qi0_center  = params.qi0_center;
plume_sigma_y = params.plume_sigma_y;
K0          = params.K0;
N_ref       = params.N_ref;
S_ref       = params.S_ref;
kN          = params.kN;
kS          = params.kS;
tau_micro   = params.tau_micro;
threshold_frac = params.threshold_frac;

% Grid
dy = Ly / (Ny - 1);
y  = linspace(-Ly/2, Ly/2, Ny).';

% Effective diffusivity K(N,S)
K = K0 * (1 + kS * abs(S)/S_ref) .* (1 + kN * (N_ref / max(N, 1e-4)));

% Time step from diffusion stability
dt = CFL * dy^2 / K;
Nt = ceil(t_end / dt);
dt = t_end / Nt;           % adjust so we end exactly at t_end
time_vec = (0:Nt). ' * dt;

% Initial plume (Gaussian)
qi = qi0_center * exp(-0.5 * (y / plume_sigma_y).^2);
qi_initial = qi;

center_idx = round((Ny + 1) / 2);
qi_initial_center = qi(center_idx);

% Storage for diagnostics

```

```

if doOutput
    qi_center      = zeros(Nt+1, 1);
    qi_total_series= zeros(Nt+1, 1);
    width_series   = zeros(Nt+1, 1);

    qi_center(1) = qi(center_idx);
    qi_total = sum(qi) * dy;
    qi_total_series(1) = qi_total;

    if qi_total > 0
        y_mean = sum(y .* qi) / qi_total;
        y2_mean = sum((y - y_mean).^2 .* qi) / qi_total;
        width_series(1) = sqrt(max(y2_mean, 0));
    else
        width_series(1) = 0;
    end
else
    qi_center      = [];
    qi_total_series= [];
    width_series   = [];
end

lifetime_found = false;
tau_persist    = t_end;    % default: still "visible" at t_end

% Time integration loop
for n = 1:Nt
    % Diffusion (Neumann-type boundaries)
    qL = [qi(1);    qi(1:end-1)];
    qR = [qi(2:end); qi(end)];
    d2qi = (qL - 2*qi + qR) / dy^2;

    qi = qi + dt * K * d2qi;

    % Bulk microphysics:
    % RHi < 100%: decay; RHi > 100%: weaker growth
    if RHi < 100
        decay_rate = (1 - RHi/100) / tau_micro;
        qi = qi - dt * decay_rate .* qi;
    elseif RHi > 100
        growth_rate = 0.05 * (RHi/100 - 1) / tau_micro; % gentle growth
        qi = qi + dt * growth_rate .* qi;
    end

    % Non-negativity
    qi = max(qi, 0);

    % Diagnostics

```

```

    qi_center_val = qi(center_idx);

    if doOutput
        qi_center(n+1) = qi_center_val;

        qi_total = sum(qi) * dy;
        qi_total_series(n+1) = qi_total;

        if qi_total > 0
            y_mean = sum(y .* qi) / qi_total;
            y2_mean = sum((y - y_mean).^2 .* qi) / qi_total;
            width_series(n+1) = sqrt(max(y2_mean, 0));
        else
            width_series(n+1) = 0;
        end
    end

    % Lifetime based on centerline amplitude
    if ~lifetime_found && qi_center_val < threshold_frac * qi_initial_center
        tau_persist = n * dt;
        lifetime_found = true;
        % (You could break here if you don't need the full time series)
        % break;
    end
end

% Pack outputs
result = struct();
result.tau_persist = tau_persist;
result.qi_final = qi;
result.qi_initial = qi_initial;
result.qi_center = qi_center;
result.qi_total = qi_total_series;
result.width = width_series;
result.time = time_vec;

grid = struct();
grid.y = y;
end

```

7.3 COMSOL simulations

High-fidelity plume simulations were performed using COMSOL Multiphysics[®]. Due to software licensing restrictions and the proprietary nature of finite-element model files, COMSOL model files cannot be shared directly. However, all governing equations, boundary conditions, and diagnostics used for comparison are fully described in Section 3.

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